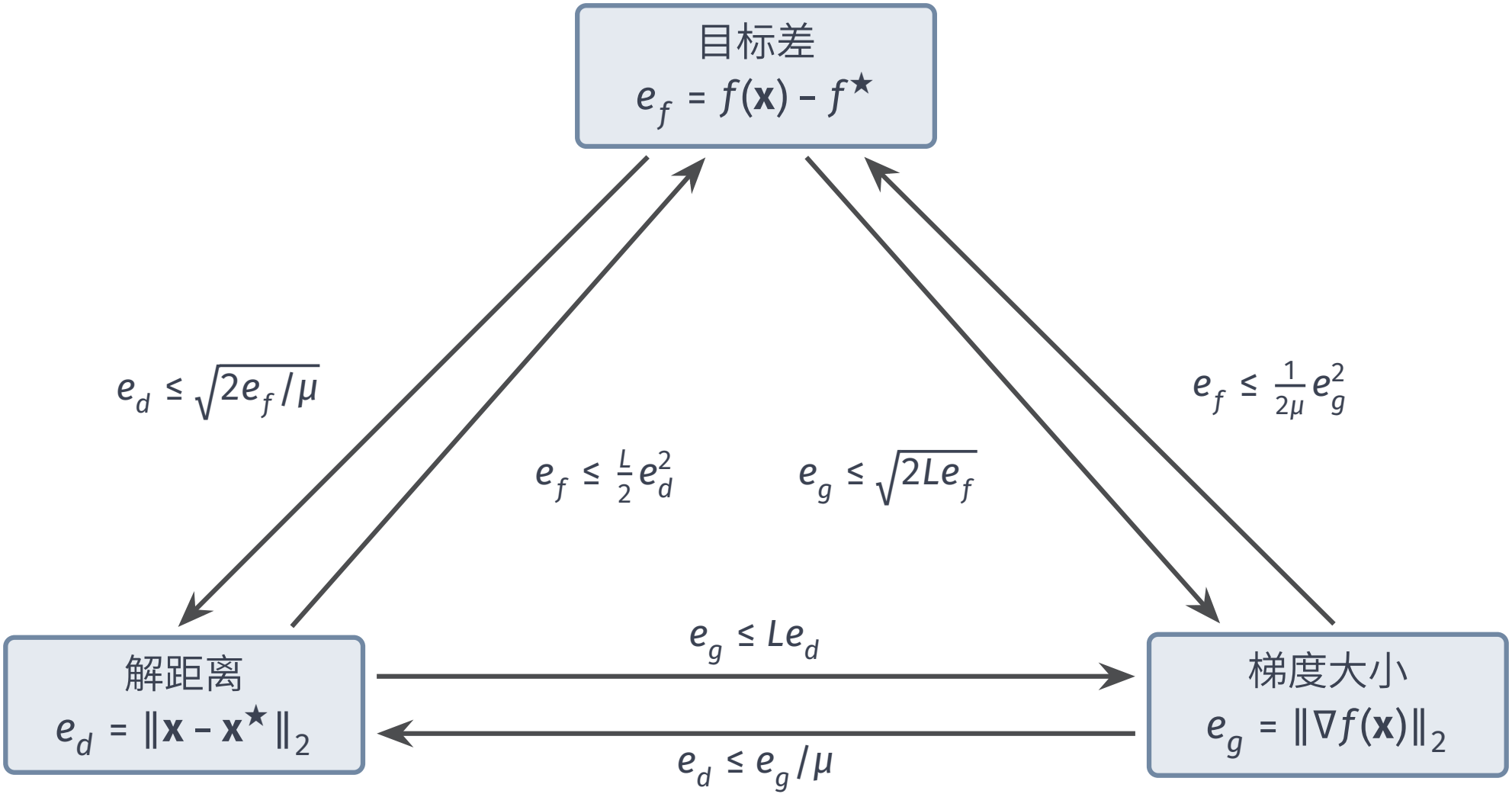
目标差
$e_f = f(\mathbf{x}) - f^\star$
解距离
$e_d = \|\mathbf{x} - \mathbf{x}^\star\|_2$
梯度大小
$e_g = \|\nabla f(\mathbf{x})\|_2$
$e_d \le \sqrt{2e_f/\mu}$
$e_f \le \frac{L}{2} e_d^2$
$e_g \le \sqrt{2Le_f}$
$e_f \le \frac{1}{2\mu} e_g^2$
$e_g \le Le_d$
$e_d \le e_g/\mu$